

Shane Kopec

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Cengage Group — Senior Full Stack Software Engineer July 2025–July 2026

- Replaced a legacy synchronization pipeline distributing updates across 170,000+ student-forked repositories spanning four GitHub organizations with an on-demand content delivery architecture.
- Led migration of a core product codebase from JavaScript to TypeScript, improving maintainability, type safety, and developer velocity across front-end and back-end services.
- Developed and maintained React/TypeScript front-end systems and Node.js services powering hundreds of thousands of GitHub Codespaces development environments, including containerized infrastructure using Docker.
- Architected validation pipeline migration from GitHub Actions to in-Codespace test execution, reducing feedback latency by ~85% (45–60s → 5–10s) and cutting infrastructure overhead.
- Implemented checksum-based authenticity validation to automatically detect and reconcile tampered exercise tests against source-of-truth repositories, ensuring content integrity and enabling live updates.
- Resolve escalated customer-facing platform issues across VS Code extensions, GitHub Codespaces environments, and repository automation workflows, maintaining service reliability for 100K+ of active environments.

Cengage Group — Junior Full Stack Software Engineer July 2023–Jul 2025

- Built and maintained 8 language-specific production Docker/devcontainer images powering GitHub Codespaces environments serving up to 30,000 student and professor seats per semester with 48,000+ provisions to date.
- Optimized GitHub Codespaces startup performance and managed CI/CD publishing pipelines using GitHub Actions and GHCR.
- Migrated computer science curricula from a legacy platform to GitHub Codespaces, enabling 24 language courses and 170,000+ students to leverage cloud-based development environments.
- Designed and developed an internal QA automation platform validating programming exercise states across GitHub Enterprise Cloud repositories, supporting 24 products and hundreds of exercises per release.
- Implemented CI/CD workflows enabling scalable QA automation across production deployments, eliminating manual testing overhead and accelerating release cadence.

Personal Project - MCP Dashboard

- Built a containerized observability platform for Model Context Protocol servers, enabling inspection, replay, and debugging of JSON-RPC tool interactions across AI workflows.
- Developed an interactive React dashboard for traffic inspection, request replay, and tool testing against connected MCP servers.

Skills: TypeScript, JavaScript, Python, React, Node.js, Express, REST APIs, VS Code Extension API, Docker, devcontainers, GitHub Codespaces, GitHub Actions, GHCR, CI/CD, Linux, SQLite, MCP

Education: Rowan University — B.S. Computer Science